

Specifications

3D AI People Counter

HX-CCD21 (AI Version)

Powered by high-precision 3D imaging, Edge AI and ReID, it delivers up to 99% accuracy. Automatically filters employees and other non-customer traffic to provide verified, effective foot traffic data you can trust.

Product Introduction

The HX-CCD21 binocular customer flow statistics terminal (AI version) employs high-precision 3D imaging technology, capable of real-time counting of personnel, with an accuracy of up to 99%. With built-in AI algorithms and edge AI + ReID full-body feature matching, it can automatically filter out non-customer noise such as employees, delivery personnel, and couriers, outputting real and valid customer flow.

Core Differentiation: Valid Customer Flow

The HX-CCD21 is not just a customer flow counter. Through edge AI + ReID full-body feature matching, it can automatically filter out non-customer noise such as employees, delivery personnel, and couriers, providing real and valid customer flow—something traditional customer flow statistics devices cannot achieve.

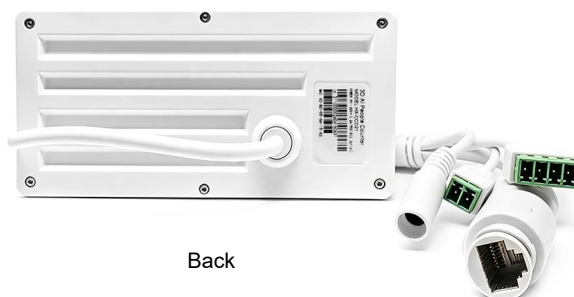
Measured data: A certain snack chain store recorded 347 visits in a day using traditional statistics, with only 143 being real customers, resulting in a noise rate of 58.8%.

Product Structure

• Product Images

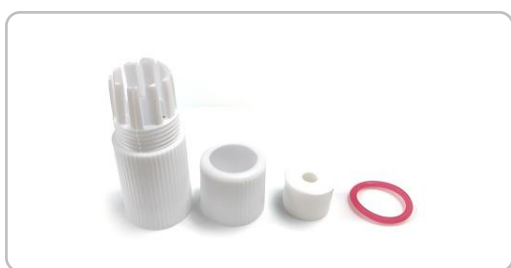


Front View



Back

• Product Accessories



Waterproof Cap



Screw Package



Installation Bracket



Instruction Manual

! If any of the above items are damaged or lost, please contact your supplier promptly.

Product Features

2 TOPS Heterogeneous Computing Power

Intelligent dual-core architecture, 20% energy efficiency improvement, real-time AI inference speed reaches millisecond-level response, no additional edge box required.

ReID Deduplication Engine

Automatically removes duplicate counts of the same person entering or staying multiple times in one day through a deep learning model of human features, with a deduplication accuracy of 80%.

99% Counting Accuracy

Real-time accurate identification of human bodies, accurately filtering out children and other large objects, with an accuracy rate of up to 99%.

Employee/Delivery/Rider Identification

Intelligent identification through employee badges accurately filters employee entry and exit data while automatically recognizing non-customer noise such as delivery personnel and couriers, avoiding data interference, with an accuracy rate of 80%.

Crowd Attribute Analysis

Gender recognition accuracy rate of 95%, age recognition accuracy rate of 80%, generating rich crowd profiles and supporting in-depth data analysis.

Full-link Privacy Compliance

No extraction of facial or other attribute information, all calculations are done locally, in compliance with personal data protection laws, ensuring user privacy.

Stay Duration Analysis

Based on ReID data, records the arrival and departure times of each valid customer, accurately calculating the duration of stay in the store, allowing for classification and analysis of customers based on different stay durations, providing insights into customer behavior characteristics.

Multi-Protocol Upload · Resume Upload After Disconnection

Supports HTTP/HTTPS/FTP/SFTP/Modbus-RTU/MQTT protocols for uploads, local computing without a server, Flash offline storage for 90 days, resume transmission after disconnection, PoE power supply, wired/wireless dual connection.

Rich Open Interfaces

The device locally opens rich application interfaces, supports RS-485 expansion, allowing developers to integrate and develop flexibly and quickly.

Multi-device Linkage and Deduplication

Supports networking and linkage of multiple devices, unified deduplication across entrances and regions, avoiding duplicate counting at multiple entry points, ensuring accurate and non-inflated customer flow data for the entire store.

Product Parameters

Sensor Parameters	
Product name	Binocular Customer Flow Statistics Terminal
Product model	HX-CCD21
Image sensor	13" Progressive Scan CMOS
Image Resolution	1920 x 1080
Pixel Size	2.7um x 2.7um
Image Output Format	RAW RGB
Lens	1mm (Fisheye Lens)
Focal Length Adjustment	Software Zoom
Field of view	Horizontal 140°, Vertical 120°
Frame rate	25fps1080
Functional Parameters	
Counting accuracy	Up to 99%
Installation height range	1.9~6.0m
Coverage range	1.1~8.4m
Height Filter	0.5~1.2m
AI Function Parameters	
Human Attribute Recognition	Gender 95%/Age 80%/Employee 80%/Deduplication 80%
AI Function Expansion	On-Demand Assessment
Recommended Height for AI Functions	2.5~4.0m
Technical Parameters	
Power	3.2-3.6W
Addressing Method	Static IP/DHCP
Offline Cache	90 Days
Data Expansion Interface	RS-485x1/6V-24V(IO Input x1, IO Output x1)
Network	Cat 5e Cable/WiFi 2.4GHZ
Power Supply Method	PoE(802.3af/at) / DC 9V-36V
Data Upload Method	HTTP POST/HTTPS POST/FTP/SFTP / Modbus-RTU / MQTT
Operating Environment	
IP Rating	IP66
Operating Temperature	-10°C-50°C

Working Humidity	20%-80%
Storage Temperature	-20°C-60°C
Storage Humidity	20%-80%
Packing Parameters	
Product Specifications	143 x 70 x 40 mm, 0.42 kg/unit
Packing Specifications	240 x 110 x 100 mm, 0.7 kg/unit (including accessories)
Outer Box Specifications	500x500x250 mm, 20 units/box, 14 kg/box
Installation Method	Ceiling-mounted/Suspension/Embedded Installation
Certification	
Product Certification	CE/FCC/ROHS/IP66 Waterproof Report/Quality Inspection Report/High and Low Temperature Test Report/Appearance Patent/Utility Model/MTBF Report/BIS Certification/Shock and Vibration Test Report

Installation Height

Installation Height (m)	Coverage Width (m)
1.9	1.1
2.5	3.7
3.0	5.2
3.5	6.4
4.0	7.3
4.5	8.0
5~6	8.4

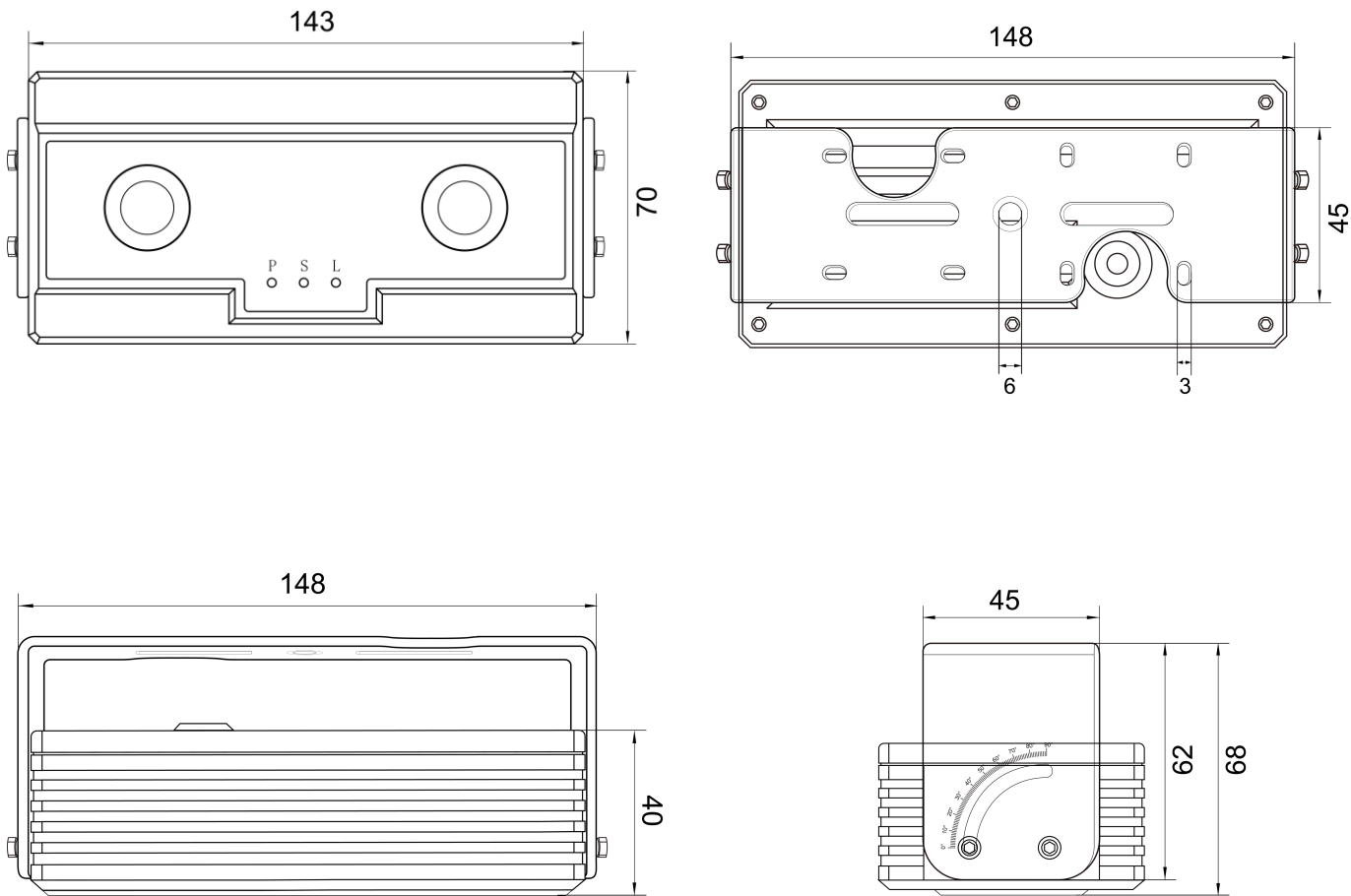
Installation Recommendations

When using AI features (ReID deduplication / gender and age recognition / employee identification), it is recommended to install at a height of 2.5~4.0m for optimal recognition results. The coverage width can reach 3.7~7.3m, meeting the width requirements of most store entrances.

• Bracket Size



Mounting Bracket (Standard Configuration)





Embedded Mounting Bracket (Optional)

